ACC250: Intro to Financial Accounting

Ch8. Accounts Receivable and Bad Debt Expense

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Central Michigan University

Learning Objectives

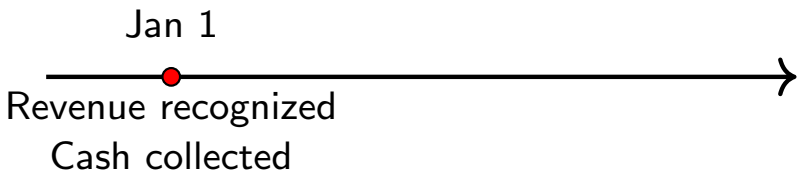
- Weigh the benefits and costs of **selling on credit**.
- Compare the **direct write-off method** and the **allowance method**.
- Record **bad debt expense**, the **allowance account**, and **write-offs**.
- Estimate bad debt expense by the **percentage of credit sales** method.

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   - Method 1. Direct Write-off Method (not allowed under GAAP)
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   - Method 2. Aging A/R — Appendix (Not for exam)

Cash Sales vs. Credit Sales[1]

Cash Sales



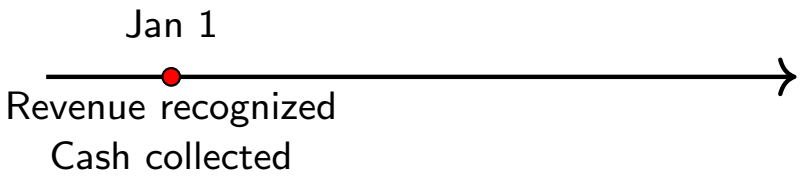
- Immediate cash inflow
- No collection risk

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Inventory		XXX

Cash Sales vs. Credit Sales[1]

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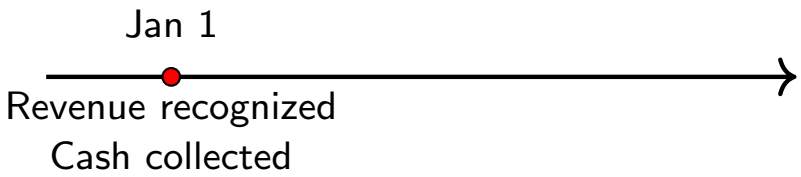
Cash	5K
Revenues	5K

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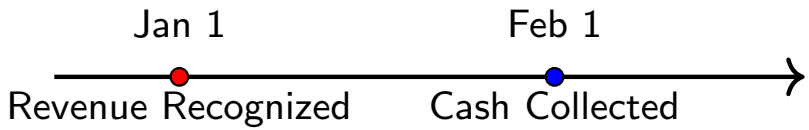
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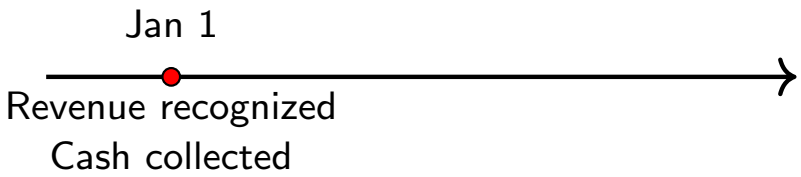
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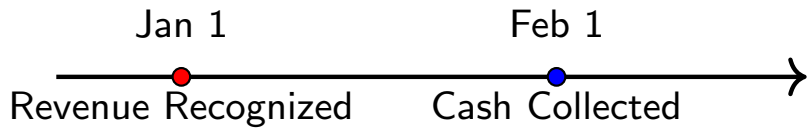
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Extending Credit: Concept, Pros, and Cons

- Companies extend credit to customers to increase sales.
  (e.g., 30-day credit terms)

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- Bad debt losses
- Cash flow delays

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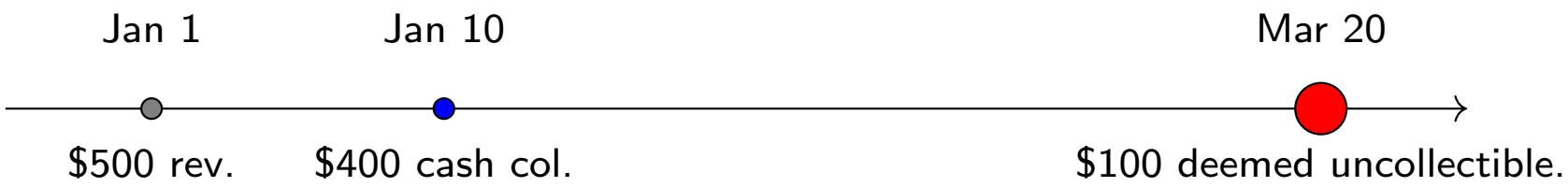
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Amounts customers owe you for products and services sold on credit.

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2,000	
EB. 4,000	

A/R for B

5,000	
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EB. 7,000	

A/R for C

11,000	2,000
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- **Collection risk exists.**

Gross and Net A/R[2]

[2] Conceptually similar to Net Sales (Ch6) and Net PPE (Ch9).
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- **Gross A/R**: The total amount owed to a company by its customers for goods/services sold on credit.

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Gross and Net A/R[2]

- **Gross A/R**: The total amount owed to a company by its customers for goods/services sold on credit.
- **Net A/R**: The amount of A/R a company expects to actually collect **after adjusting uncollectible A/R**.

EX. Gross Accounts Receivable: \$20K but we expect that \$1K is uncollectible.

Gross A/R	\$20,000
Less: Allowance for Doubtful Accounts (xA)[3]	\$1,000
Net A/R	\$19,000

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- For each customer, collection risk exists.
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  - Some may pay less than the amount owed.
- This uncollectible portion is called **bad debt**.
- Bad debt should be **recognized** as expense in a **proper period**.

How much? and When?

Bad Debt Expense

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Operating expense resulting from credit sales that become uncollectible.

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- Usually included in SG&A Expense.[a]
- Doesn't affect gross profit.
- Affects **operating income** and **net income**.

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Simplified Income Statement

Revenue	$5,000
CGS	3,000
Gross Profit	$2,000
SG&A Expense	800
Operating Income	1,200
Interest Expense	150
Marketing Expense	50
Pretax Income	1,000
Income Tax (20%)	200
Net Income	$800

Bad Debt Expense

Balance Sheet Equation

Assets = Liabilities + Shareholders' Equity

Higher Bad Debt Expense

1. Higher SG&A Expense
2. Lower Operating Income
3. Lower Net Income
4. Lower Retained Earnings
5. Lower Shareholder's Equity

Higher Bad Debt Expense

1. Lower Net A/R

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Method 1. Direct Write-off Method

- Recognizes bad debt expense when a specific account is deemed uncollectible.
- No estimate of uncollectible accounts is made.
- Not allowed under GAAP for financial reporting.
- Allowed for tax purposes.

Method 1. Direct Write-off Method

EX. On Jan 1, we made \$5K in credit sales and expect that 10% of the sales is uncollectible. A customer declared bankruptcy and we wrote off \$500 on Feb 1. On Feb 10, \$4,500 was collected. Prepare journal entries:



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	A/R (-A)		500

Feb 10:	Cash (+A)	4500	
	A/R (-A)		4500

Method 1. Direct Write-off Method

Note that we do not make adjustment journal entries at the end of January under Direct Write-off Method.

Income Statement based on the given info:

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	January
Revenue	$5,000
BDE	0
Net Income	$5,000

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Revenue	$0
BDE	**500**
Net Income	-$500

Matching Principle

- Expenses should be recognized **in the same period as the revenues** they help to generate, regardless of when cash is paid. In other words,
- Expenses should be recognized when the resources are **used up** to generate the revenues.

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Method 2. Allowance Method

To comply with the matching principle: At the end of each accounting period,

- **BDE** is estimated and recorded.
- **A/R, net** is adjusted to reflect the estimated uncollectible amount.
  $\rightarrow$ **AJE** is needed.[4]

[4] AJE: Adjusting Journal Entry

Method 2. Allowance Method

The impact of the AJE:

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- Increases BDE (expense) on the income statement.
- Decreases Net, A/R (asset) on the balance sheet.
- Decreases net income and total assets.

Method 2. Allowance Method

Allowance for Doubtful Accounts (ADA, Contra Asset, xA)

The estimated amount of uncollectible A/R.

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Allowance for Doubtful Accounts (ADA, Contra Asset, xA)

The estimated amount of uncollectible A/R.

- A contra-asset account (xA), so it has a credit balance.
- **Net A/R= Gross A/R - ADA**

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	ADA (+xA)		500

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	ADA (+xA)		500

A/R Reporting:[a]

Gross A/R:	\$5,000
Less: ADA:	\$500
Net A/R:	**\$4,500**

[a] This is the way A/R is reported on the balance sheet.

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Remember!

- Customers who do not pay will be identified later (Feb).
- **Bad debt expense** is recognized in the same period as the credit sales (Jan).

Method 2. Allowance Method

Write-off of A/R: When a specific A/R is deemed uncollectible:

- Write off the A/R against the allowance account.

Allowance for Doubtful Accounts (-xA)	500	
Accounts Receivable (-A)		500

- No additional expenses are recorded at this time as it was already recognized as BDE in Jan.

- **Net A/R is not affected by the write-off.**

Gross A/R	\$5,000 - X
ADA (xA)	\$500 - X
Net A/R	**\$4,500**[5]

[5] (5000 - X) - (500 - X)= 4500.

Method 2. Allowance Method

ADA is akin to ADE.[6]

- Both are contra-asset accounts.
- Both reduce the book value of their related assets (A/R and PPE, respectively)

[6] ADA: Allowance for Doubtful Accounts, ADE: Accumulated Depreciation Expense

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A/R and ADA

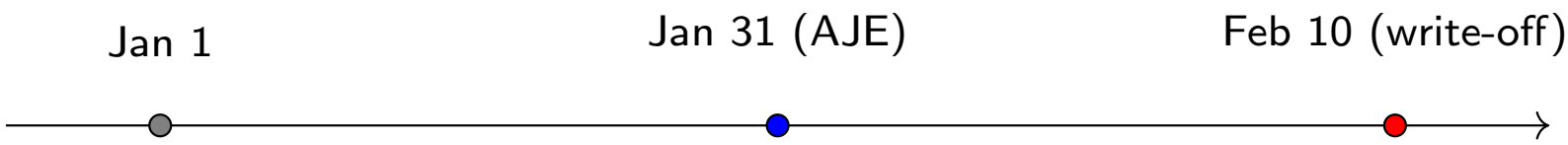
Description	Amount
Gross A/R	$5,000
Less: ADA (xA)	(500)
Net A/R	$4,500

PPE and ADE

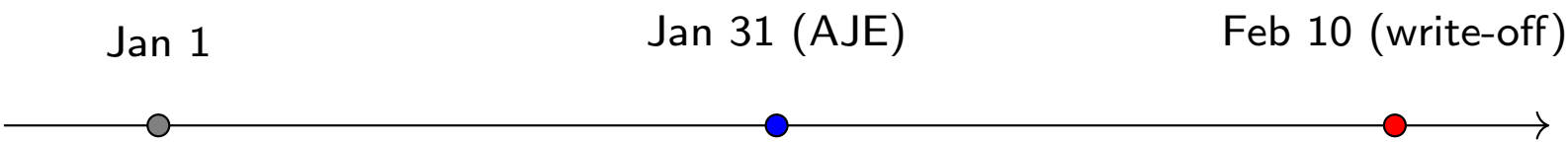
Description	Amount
Gross PPE	$10,000
Less: ADE (xA)	(2,000)
Net PPE	$8,000

[6] ADA: Allowance for Doubtful Accounts, ADE: Accumulated Depreciation Expense

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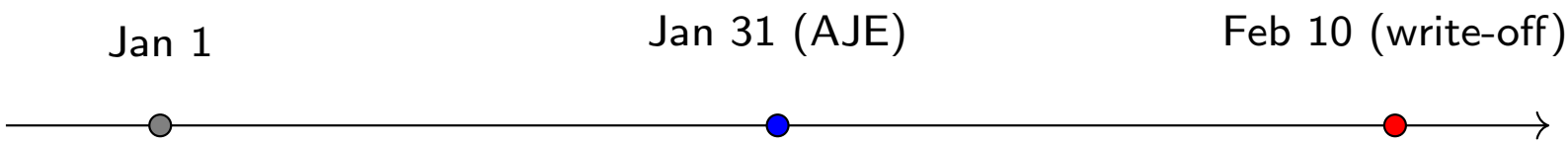
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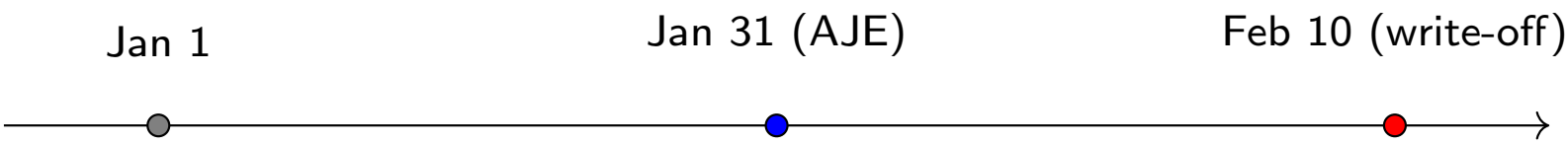
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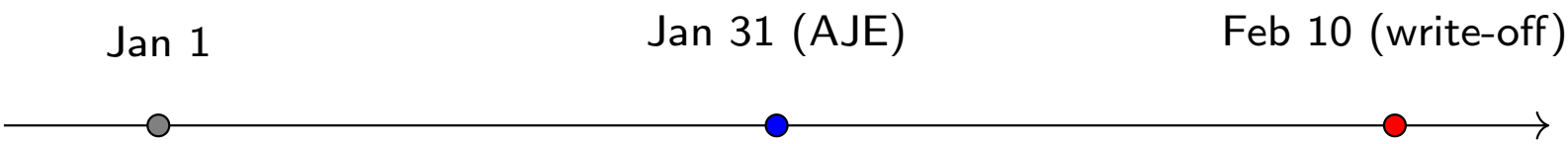
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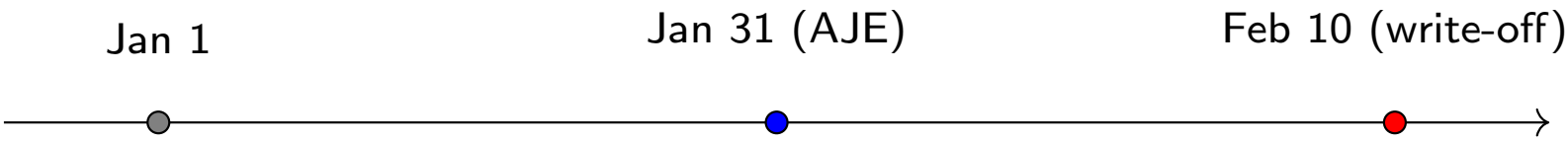
Jan 31:	BDE (+E)	500	
	ADA (+xA)		500

Feb 10:	ADA (-xA)	500	
	A/R (-A)		500

A/R before and after write-off

A/R as of Jan 31	
Gross A/R	\$5,000
Less: ADA (xA)	\$500
Net A/R	**\$4,500**

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Feb 10:	ADA (-xA)	500	
	A/R (-A)		500

A/R before and after write-off

A/R as of Jan 31	
Gross A/R	\$5,000
Less: ADA (xA)	\$500
Net A/R	**\$4,500**

$\rightarrow$

A/R as of Feb 10	
Gross A/R	\$4,500
Less: ADA (xA)	\$0
Net A/R	**\$4,500**

Note: Financial Statement Example: Crocs, Inc.

Figure: Balance Sheet

CROCS, INC. AND SUBSIDIARIES
CONSOLIDATED BALANCE SHEETS
(in thousands, except share and par value amounts)

	December 31,	
	2024	**2023**
ASSETS		
Current assets:		
Cash and cash equivalents	$ 180,485	$ 149,288
Restricted cash — current	—	2
Accounts receivable, net of allowances of $31,579 and $27,591, respectively	257,657	305,747

Note: Financial Statement Example: Crocs, Inc.

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Figure: Disclosures in Note

Accounts Receivable, Net

Accounts receivable are recorded at invoiced amounts, net of reserves and allowances. We reduce the carrying value for estimated uncollectible accounts based on a variety of factors including the length of time receivables are past due, economic trends and conditions affecting our customer base, and historical collection experience. Specific provisions are recorded for individual receivables when we become aware of a customer's inability to meet its financial obligations. We write off accounts receivable to the reserves when they are deemed uncollectible or, in certain jurisdictions, when legally able to do so. See Schedule II in Item 15. *Exhibits, Financial Statement Schedule* of this Annual Report on Form 10-K for more information.

Note: Bad Debt Exp. vs Allowance for Doubtful Acc.

Allowance for Doubtful Accounts

Bad Debt Expense

Note: Bad Debt Exp. vs Allowance for Doubtful Acc.

Allowance for Doubtful Accounts

- Captures the outstanding balance of uncollectible A/R

Bad Debt Expense

- Captures how much adjustment is needed to ADA

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- Contra-asset account

Bad Debt Expense

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- Expense account

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Allowance for Doubtful Accounts

- Captures the outstanding balance of uncollectible A/R
- Contra-asset account
- Balance sheet account

Bad Debt Expense

- Captures how much adjustment is needed to ADA
- Expense account
- Income statement account

Relationship between ADA and BDE

Allowance for Doubtful Accounts (ADA, Contra Asset, xA)

The estimated amount of uncollectible A/R.

Relationship between ADA and BDE

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The estimated amount of uncollectible A/R.

Bad Debt Expense (BDE, Expense)

Operating expense resulting from credit sales that become uncollectible.

Relationship between ADA and BDE

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The estimated amount of uncollectible A/R.

Bad Debt Expense (BDE, Expense)

Operating expense resulting from credit sales that become uncollectible.

To get the ending balance of ADA:

- Start with the beginning balance of ADA.
- Add the BDE for the period. (ADA increases)
- Subtract the Write-off for the period. (ADA decreases)

Relationship between ADA and BDE

Allowance for Doubtful Accounts (ADA, Contra Asset, xA)

The estimated amount of uncollectible A/R.

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Operating expense resulting from credit sales that become uncollectible.

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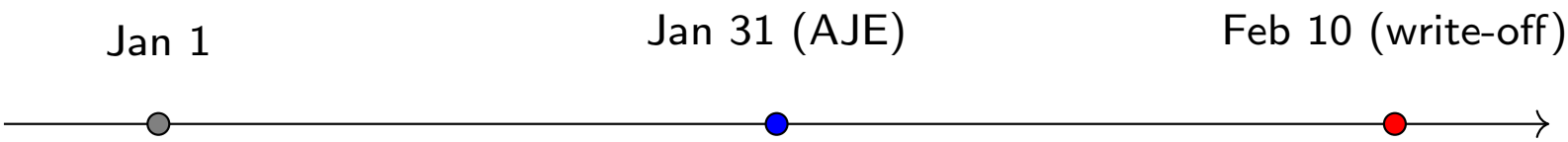
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Allowance for Doubtful Accounts Equation

$$\text{ADA}_{t+1} = \text{ADA}_t + \text{BDE}_{t+1} - \text{Write-off}_{t+1}$$

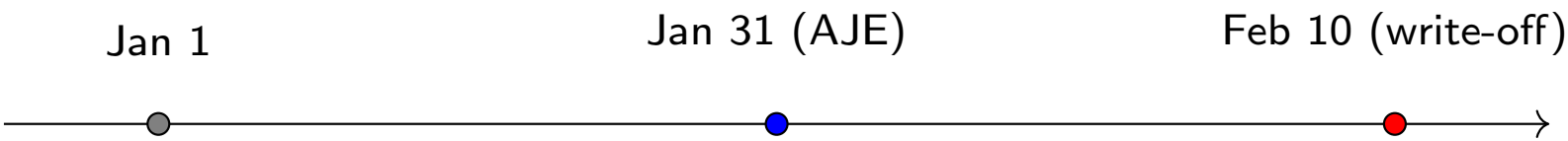
Illustration

EX. On Jan 1, a company made \$10K in credit sales. It estimates 5% of sales will be uncollectible. The cost of goods sold (CGS) for these sales is \$6K. On Feb 10, \$300 of A/R deemed uncollectible.



Illustration

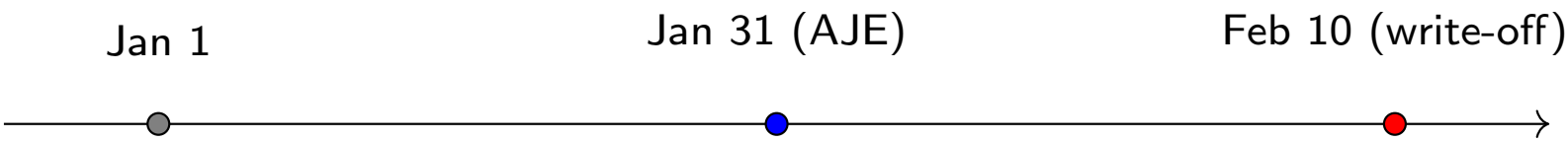
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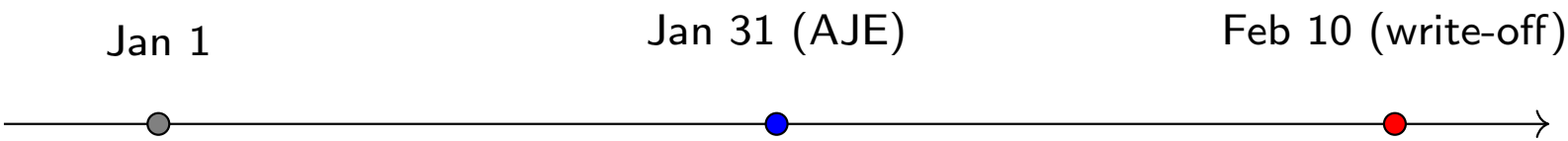


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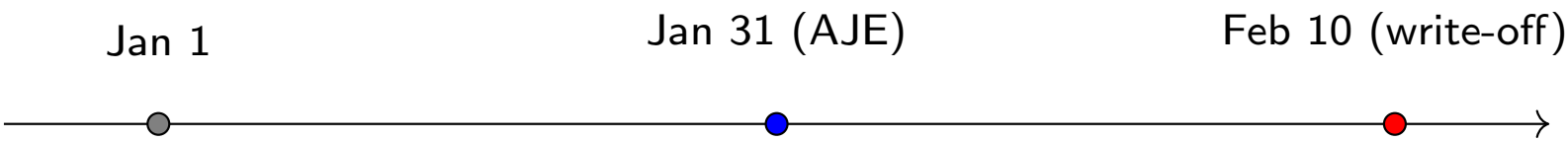
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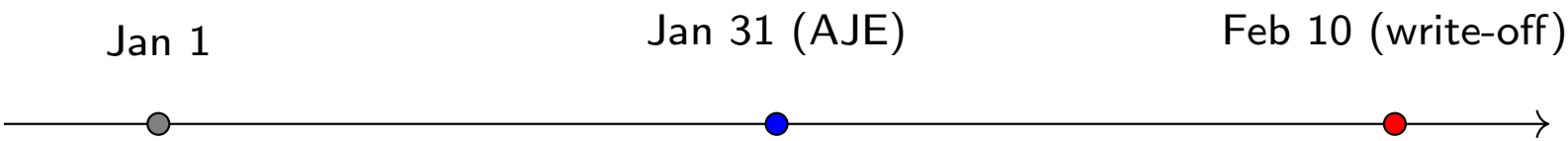
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- Step 2: Estimate Bad Debt Expense

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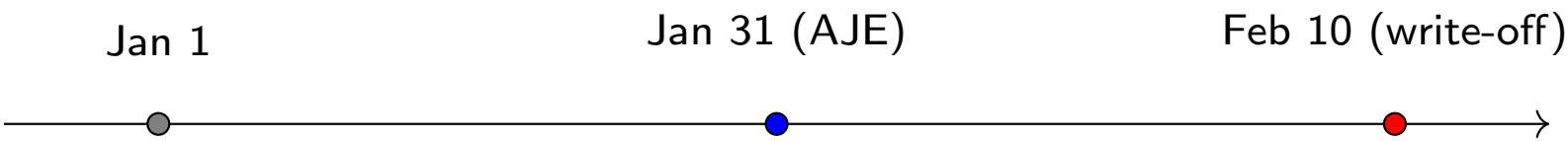
Jan 1:	CGS (+E)	6K	
	Inv (-A)		6K

- Step 2: Estimate Bad Debt Expense

Jan 31:	BDE (+E)	500	
	ADA (+xA, -A)		500

Illustration

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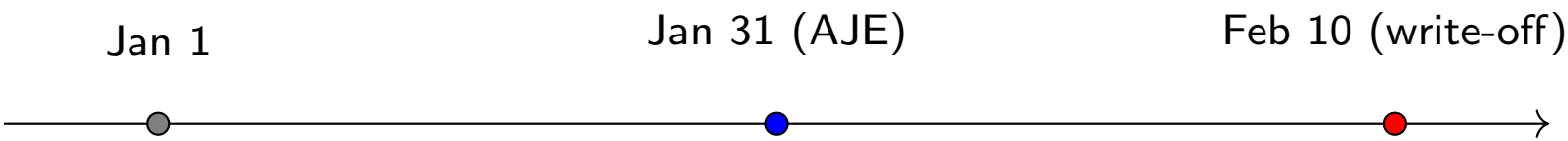
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- Step 3: Write-off of bad debts

Illustration

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	Inv (-A)		6K

- Step 2: Estimate Bad Debt Expense

Jan 31:	BDE (+E)	500	
	ADA (+xA, -A)		500

- Step 3: Write-off of bad debts

Feb 10:	ADA (-xA, +A)	300	
	A/R(-A)		300

Table: Partial Income Statement for January

Sales Revenue	$10,000
Cost of Goods Sold	$6,000
Bad Debt Expense	$500
Net Income:	$3,500

Table: Net A/R as of Jan 31

Gross A/R	$10,000
Less: ADA	$500
Net A/R	**$9,500**

Table: Net A/R as of Feb 10

Gross A/R	$9,700
Less: ADA	$200
Net A/R	**$9,500**

Note: It is possible that A/R deemed uncollectible earlier can be collected later! Accounting for this recovery of uncollectible A/R goes beyond this course.

Table of Contents

1. Credit Sales and Accounts Receivable
2. Bad Debt Expense
3. Two Methods accounting for Bad Debt Expense
   - Method 1. Direct Write-off Method (not allowed under GAAP)
   - Method 2. Allowance Method (allowed under GAAP)
4. Estimation of Bad Debt Expense
   - Method 1. % of Credit Sales
   - Method 2. Aging A/R — Appendix (Not for exam)

Estimation of Bad Debt Expense

Two methods for BDE estimation

1. Percentage of Credit Sales Method
2. Aging of Accounts Receivable

* This course uses **Method 1**. Method 2 is widely used in practice and is shown as an Appendix (not for exam).

Method 1. % of Credit Sales

- Focus: BDE
  $\rightarrow$ I/S approach.

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Method 1. % of Credit Sales

- Focus: BDE
  $\rightarrow$ I/S approach.
- BDE is estimated by multiplying the historical % of bad debt losses by the current period's credit sales.
- Then ADA is adjusted by the estimated BDE.
- Simple and intuitive, but not widely used.

EX. VFC has experienced bad debt losses of 2% of credit sales in prior periods. VFC recorded credit sales of $120K in January.

- Calculate BDE this month.

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  $120K * 2%= $2,400.

- Prepare journal entries for the recognition of Bad debt expense.

EX. VFC has experienced bad debt losses of 2% of credit sales in prior periods. VFC recorded credit sales of $120K in January.

- Calculate BDE this month.
  $120K * 2%= $2,400.
- Prepare journal entries for the recognition of Bad debt expense.

Jan 31:	BDE (+E)	2,400	
	ADA (+xA, -A)		2,400

Appendix (Not for exam): Table of Contents

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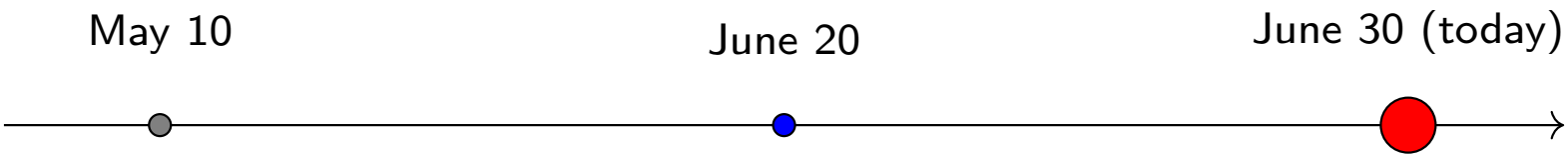
Appendix (Not for exam): Method 2. Aging A/R

- Focus: ADA
  $\rightarrow$ B/S approach.
- The ending balance of ADA is estimated considering ages of A/Rs.
- Principle: *The older an A/R is, The less likely it is to be collectible.*
- Complex, but widely used.

Appendix (Not for exam): Aging A/R: Example

EX. Our company has the following uncollected A/Rs as of June 30.

- \$5K credit sales made on May 10.
- \$5K credit sales made on June 20.



Based on the history of uncollected A/R:

- 31–60 days: 10% estimated uncollectible.
- 0–30 days: 2% estimated uncollectible.

Bad debt estimate for each group:

- May 10 sales (\$5,000, 51 days old): $10\% \times \$5K = \$500$.
- June 20 sales (\$5,000, 10 days old): $2\% \times \$5K = \$100$.

ADA at the end of Q2
$= \$500 + \$100$
$= \$600$.

Appendix (Not for exam): Aging A/R: ADA, BDE, and Journal Entries

- $600 is the updated ending balance of ADA.
- It doesn't mean that BDE for Q2 is $600.
- Recall: BDE captures how much adjustment is needed to ADA.
- Journal entries on June 30 assuming Beg.Bal of ADA was $200.

ADA (xA)

	BB. 200
	?
	EB. 600

BDE (+E)	400	
ADA (+xA, -A)		400

Summary

- At the beg of Q2, $200 of A/R was considered uncollectible.
- During Q2, there were credit sales.
- At the end of Q2, $600 of A/R is considered uncollectible.
- $400 of BDE should be recognized.

Appendix (Not for exam): Aging A/R: Illustration

Our company's aging schedule and A/R data as of Dec 31 are shown below. The beginning balance of the ADA on Jan 1 was $1K.

Aging Schedule:

Age	A/R	Est. % Uncollectible	Uncollectible Amount
0-30 days	$50,000	1%	$500
31-60 days	30,000	5%	1,500
Over 60 days	20,000	10%	2,000
Total	$100,000		$4,000

Requirements:

1. Calculate the ending balance of the ADA as of December 31.
2. Calculate the BDE for the year.
3. Complete the journal entry to record the BDE on December 31.

Appendix (Not for exam): Aging A/R: Illustration (Solution)

1. Ending balance of the ADA as of Dec 31:
   $4,000, as determined by the total uncollectible amount from the aging schedule.

2. Bad Debt Expense (BDE) for the year:
   $4K (ending ADA) - $1K (beginning ADA) = $3K.

3. Journal entry to record the BDE on Dec 31:

Dec 31:	BDE (+E)	3K	
	ADA (+xA, -A)		3K